

MATH 2790 Fall 2025

Homework 9

Due date: Wednesday, Oct 29

Instructions:

- (a) It is expected to show detailed work for every problem. You are allowed to have a discussion with peers on some challenging questions. But then you should work on your own to finish the assignment.
- (b) All questions in this assignment are from sections 4.2–4.3 and 4.6.

Problems:

- 1. Question (1)(a)(b) on page 222 of textbook under section 4.2.
- 2. Question (2) on page 222 of textbook under section 4.2.
- 3. Question (4) on page 222 of textbook under section 4.2. (Hint: use a financial calculator to solve for the annual effective interest rate)
- 4. Determine the present value of 1 payable at the end of years 7, 11, 15, 19, 23, and 27. Pick the correct choice and show your work to support your choice.

$$(A) \frac{a_{28} - a_4}{\ddot{s}_4} \quad (B) \frac{a_{28} - a_4}{s_4} \quad (C) \frac{a_{27} - a_3}{\ddot{s}_4}$$

$$(D) \frac{a_{27} - a_3}{s_4} \quad (E) \frac{a_{27} - a_3}{a_4} \quad (F) \frac{a_{28} - a_4}{a_4}$$

- 5. Question (1) on page 223 of textbook under section 4.3.
- 6. Question (4) on page 223 of textbook under section 4.3.
- 7. Question (1) on page 226 of textbook under section 4.6.
- 8. Question (2) on page 226 of textbook under section 4.6.
- 9. Question (3) on page 226 of textbook under section 4.6.